

Recommendation letter
from Associate Professor Asen Bozhilov,
Vice-dean of Faculty of Mathematics and Informatics, Sofia University
Department of Algebra
Sofia, Bulgaria

Applicant: Kristiyan Ivanov Garchev

To whom it may concern:

I am writing with great pleasure to recommend Kristiyan Garchev for admission to your graduate program. I have known Kristiyan through his work in several advanced mathematical subjects, as well as through his broader academic development as one of the strongest and most ambitious students in our faculty. In my view, he belongs to a very small group of students whose talent is immediately visible, but whose discipline and intellectual seriousness are even more impressive than their raw ability.

My direct impression of Kristiyan was formed first through his performance in core algebraic subjects such as Linear Algebra, Abstract Algebra, and Matrix Algebra. In these courses, he distinguished himself not only by mastering the formal material, but by showing genuine mathematical maturity. He is able to work comfortably with abstraction, to recognize structural similarities between different mathematical objects, and to move beyond routine problem solving toward a deeper conceptual understanding.

This impression was reinforced even more strongly in advanced subjects including Galois Theory, Commutative Algebra, Group Representation Theory, and Introduction to Homological Algebra. These are not courses in which one succeeds by memorization or technical fluency alone. They require a high degree of precision, patience, and the ability to internalize abstract definitions and arguments. Kristiyan handled this material with unusual confidence. He has the rare ability to understand algebra as a coherent language rather than as a collection of isolated topics, and this is one of the clearest signs of his promise for serious graduate study.

Kristiyan has also shown that his strengths are not confined to pure mathematics. In more quantitative and applied subjects such as Probability and Statistics, Finance, and Econometrics, he demonstrated the ability to carry the same rigor into areas where modeling, uncertainty, and interpretation matter. This is equally evident in his engagement with machine learning-related topics, where mathematical structure, statistical reasoning, and careful interpretation are just as important as technical implementation. He is able to move between abstract theory and quantitatively grounded application without losing depth in either direction.

His excellence is not limited to coursework. Kristiyan has an exceptional record in mathematical competitions, including being an International Mathematical Olympiad gold medalist, along with many other gold medals, first-place national distinctions, and strong international performances. He has also distinguished himself in competitions connected with the Bulgarian Academy of Sciences, including open-problem settings that required genuinely creative mathematical thinking. These results are important because they reveal

originality, resilience, and technical depth under pressure. They confirm what one already sees in the classroom: Kristiyan is not merely well prepared, but unusually gifted.

Another significant part of his academic profile is his work as a Teaching Assistant in more than seven subjects, including Linear Algebra, Abstract Algebra, Discrete Mathematics, and Calculus-based foundational courses, as well as Competitive Mathematics, where he actively aided the international university math team's preparation. This is a serious responsibility, especially in mathematically demanding courses where students need both technical help and conceptual clarification. Kristiyan has shown that he can explain difficult material clearly and responsibly. A student who teaches effectively in these subjects has already moved beyond personal success and begun to function as a true member of the academic community.

A series of discussions with him on computational mathematics have strengthened my view of his capacity for abstract reasoning, technical engagement, and genuine potential to do serious advanced work at the interface of mathematics and computation. Furthermore, I should also mention that Kristiyan's potential extends well beyond pure mathematics.

He has cultivated substantial interests in machine learning, quantitative finance, and bioinformatics, as reflected in the numerous courses he had taken in these areas. He brings a mathematician's way of thinking into each of them. This broader engagement is also evident in his participation in the PCI Summer School for ML Scientific Research, a selective and highly regarded program that brings together students with strong potential for research in machine learning and related quantitative fields. He also developed a research-oriented machine learning project related to RNA residue sequence 3D folding prediction, which further illustrates both his mathematical discipline and his ability to work effectively on technically sophisticated interdisciplinary problems.

Even outside the formal curriculum, his interests reflect the same cast of mind. From a young age, he has had a longstanding fascination with solving Rubik's cubes, and since then he has obtained an impressive collection of over 100 variants. This alone is strong evidence of how naturally he is drawn to structure, transformation, and nontrivial systems. As his interest deepened, he came to understand the underlying mathematical ideas behind them, especially those related to commutators. In his own words, once he had understood one simple principle, he was able to invent moves to solve them all. In my opinion, hobbies like this, however playful and silly they may appear on the surface, when taken to such an extent, are the pinnacles that showcase genuine pursuit of complexity, underlying principles, abstraction, and intellectual challenge.

In summary, I recommend Kristiyan Garchev without reservation and with the highest enthusiasm. He is an unusually strong young mathematician with broad quantitative preparation, outstanding competition credentials, proven teaching ability, and the intellectual ambition required for serious graduate study. I am confident that he will thrive in a demanding academic environment and fully justify your confidence in him.

Sincerely,
Assoc. Prof. Bojilov
Vice Dean of Sofia University "St. Kliment Ohridski"

